Script-Controlled ACL – Restrict Record Access Based on Field Value Project

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Semester: 5

Script-Controlled ACL – Restrict Record Access Based on Field Value Project

Introduction :

The Script-Controlled ACL – Restrict Record Access Based on Field Value project focuses on improving data security in ServiceNow by controlling user access to records based on specific field values. An Access Control List (ACL) determines whether a user can view, create, update, or delete a particular record or field. In this project, a script is added to the ACL to dynamically check the value of a field before granting access. This provides flexible and secure access control, ensuring that users can access only the records they are authorized to view or modify.

Project Objective :

The main objective of this project is to implement dynamic record-level security in ServiceNow using Script-Controlled ACLs.

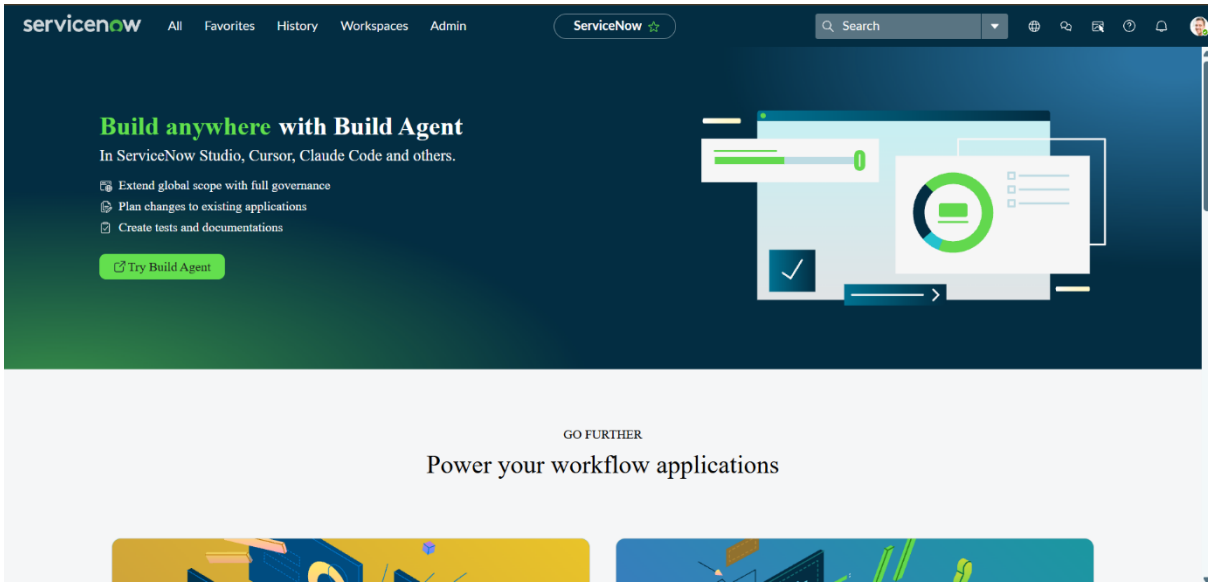
The project aims to:

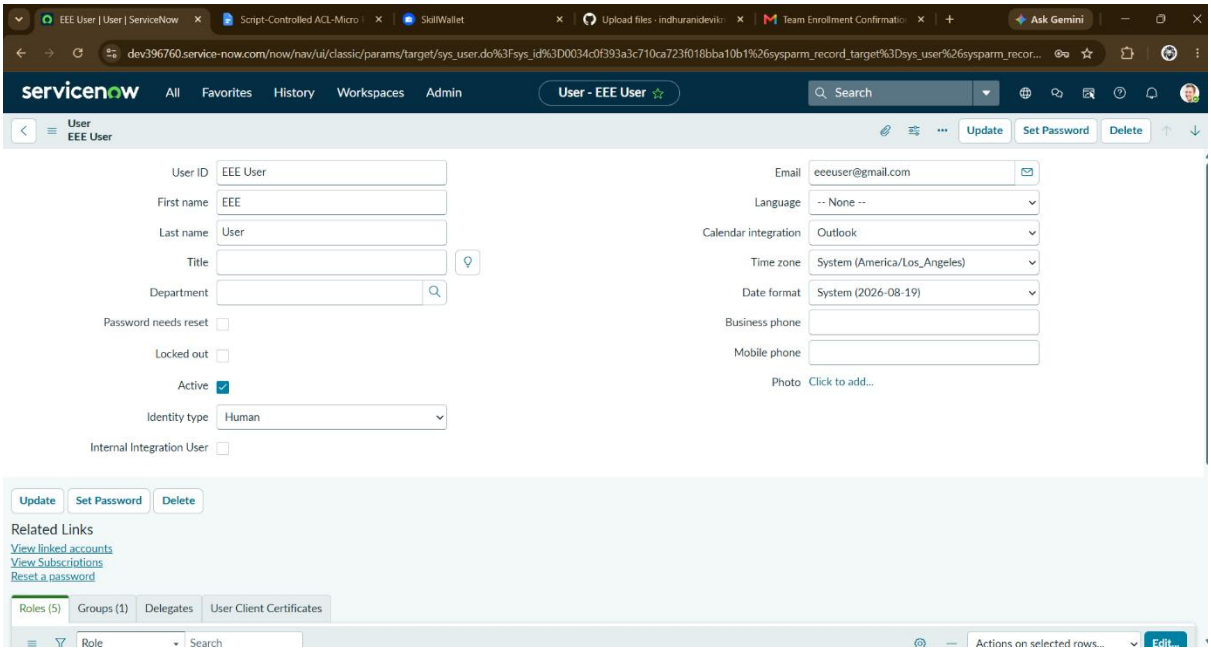
- Restrict access to records based on the value of a specific field.
- Allow authorized users to view or modify records according to defined conditions.
- Prevent unauthorized users from accessing restricted information.
- Demonstrate the use of ACLs and server-side scripting for dynamic access control.
- Improve data security and maintain proper authorization within the ServiceNow platform.

Creation of Users and Roles

User Creation:

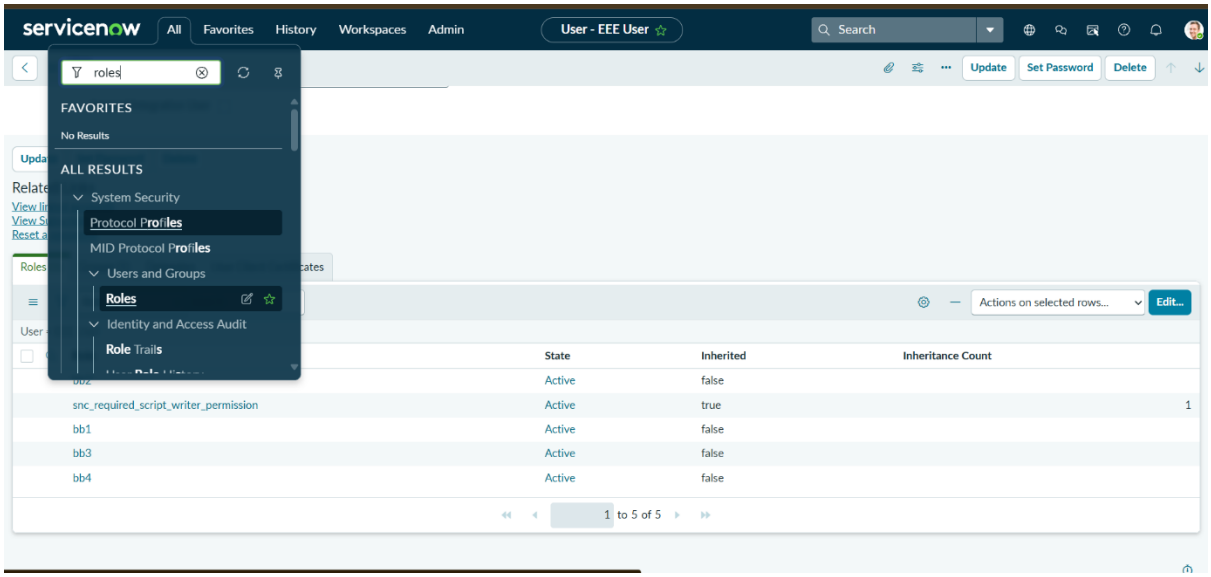
1. Log in to the ServiceNow instance with admin access.
2. Create a test user:
   - Navigate to User Administration → Users
   - Click New
   - User ID: EEE User
   - First name: EEE
   - Last name: User
   - Email: eeeuser@gmail.com
   - Save the record

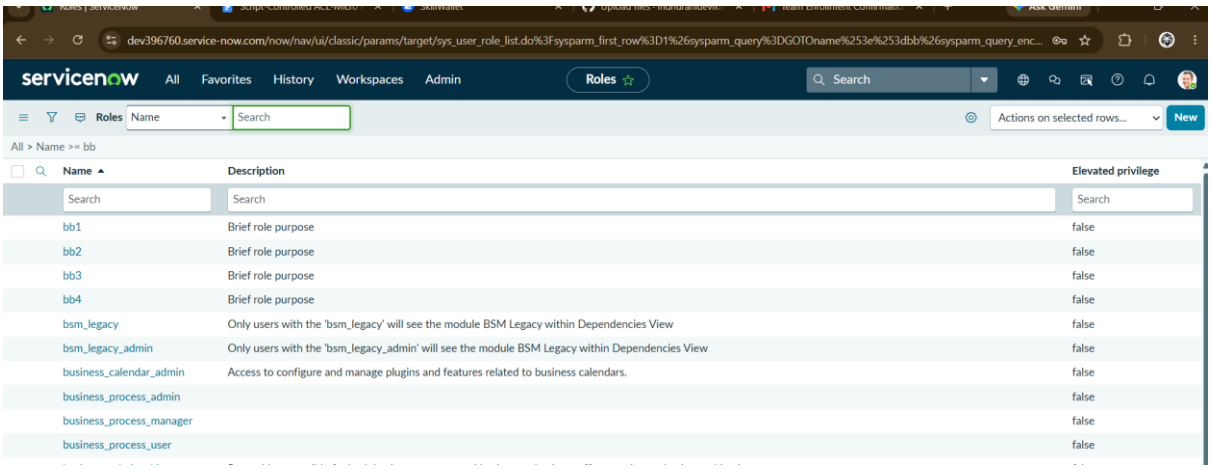


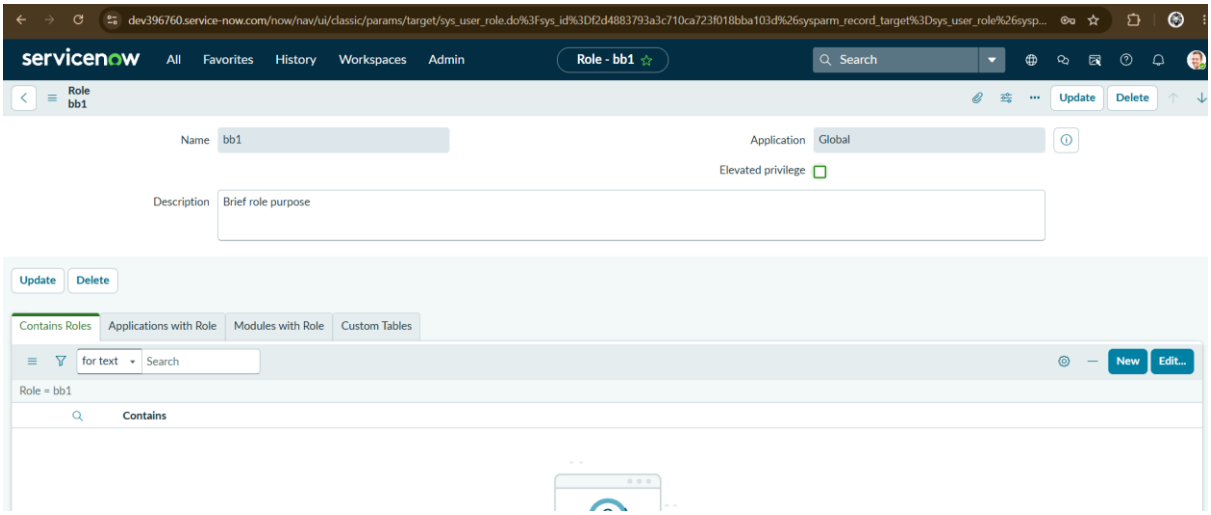


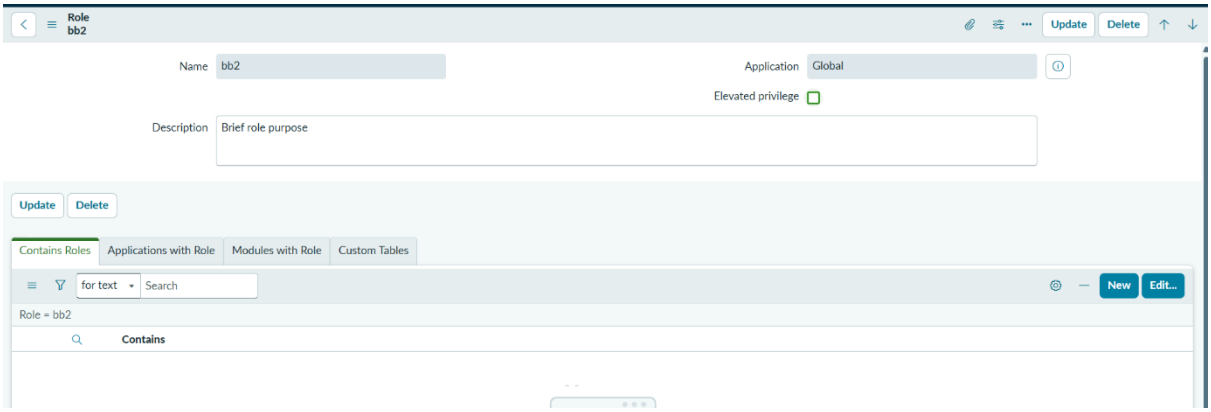
Role Creation

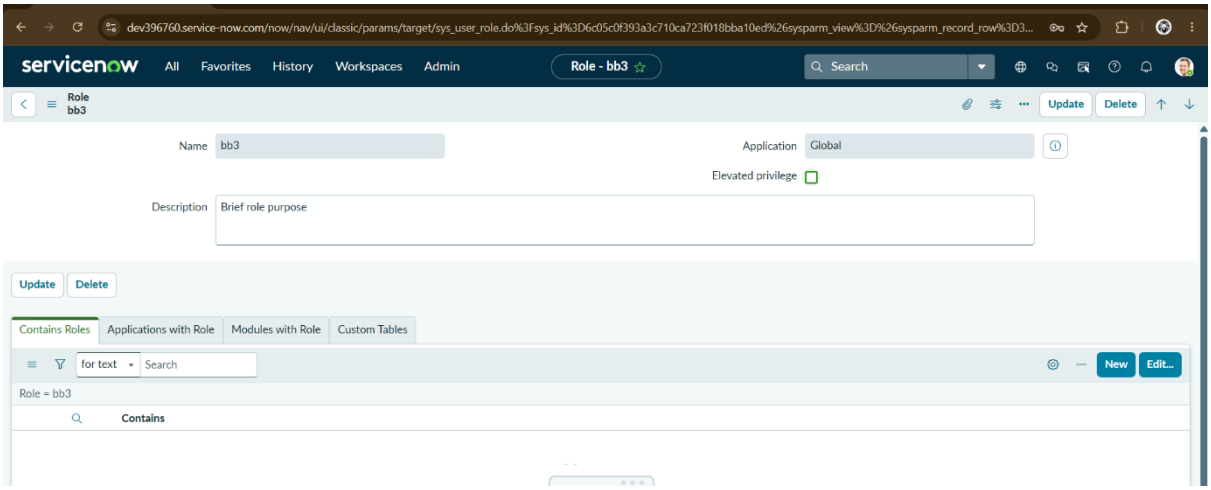
- Navigate to **User Administration → Roles**
- Click **New**
- **Name**: Enter role name as bb1
- **Description**: Brief role purpose
- Click **Submit**
- Create other roles also like **bb2,bb3,bb4.(follow the same procedure)**
- Assign all the roles to the EEE User

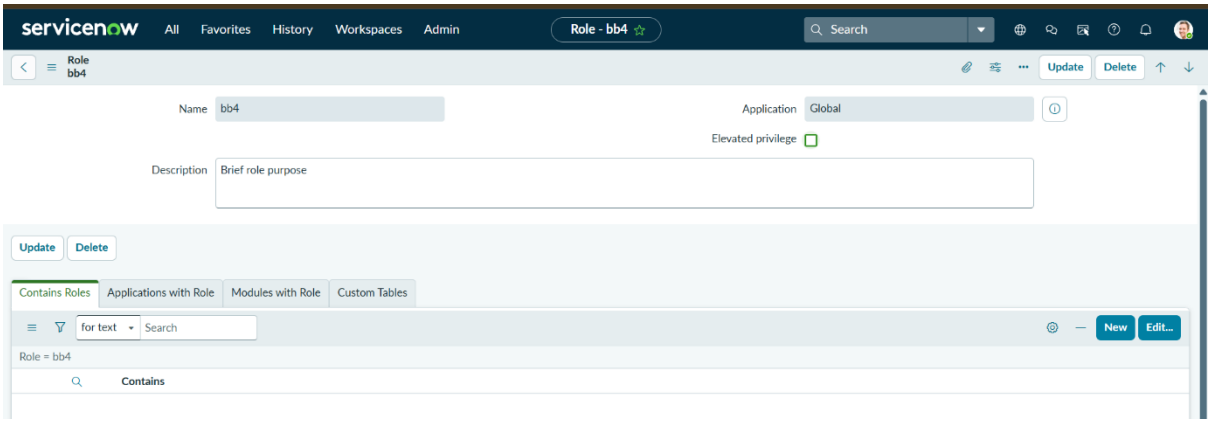

servicenow
All
Favorites
History
Workspaces
Admin
User - EEE User
Search
Update
Set Password
Delete
roles
FAVORITES
No Results
ALL RESULTS
System Security
Protocol Profiles
MID Protocol Profiles
Users and Groups
Roles
Identity and Access Audit
Role Trails
Actions on selected rows...
Edit...
State
Inherited
Inheritance Count
Active
false
snc_required_script_writer_permission
Active
true
1
bb1
Active
false
bb3
Active
false
bb4
Active
false
1 to 5 of 5

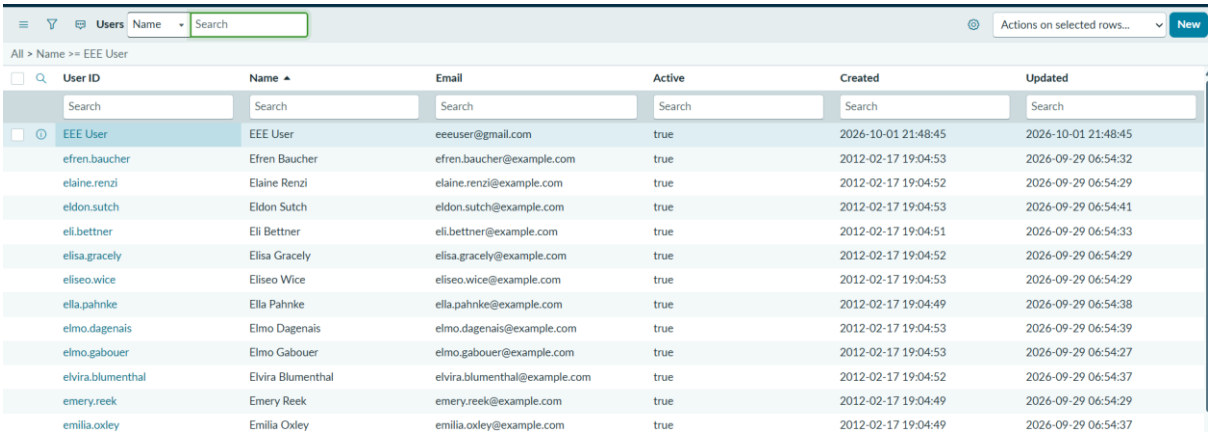

dev396760.service-now.com/now/nav/ui/classic/params/target/sys_user_role_list.do%3Fsysparm_first_row%3D1%26sysparm_query%3DGOTOname%253e%253dbb%26sysparm_query_enc...
servicenow
All
Favorites
History
Workspaces
Admin
Roles
Search
Roles
Name
Search
Actions on selected rows...
New
All > Name >= bb
Name
Description
Elevated privilege
Search
bb1
Brief role purpose
false
bb2
Brief role purpose
false
bb3
Brief role purpose
false
bb4
Brief role purpose
false
bsm_legacy
Only users with the 'bsm_legacy' will see the module BSM Legacy within Dependencies View
false
bsm_legacy_admin
Only users with the 'bsm_legacy_admin' will see the module BSM Legacy within Dependencies View
false
business_calendar_admin
Access to configure and manage plugins and features related to business calendars.
false
business_process_admin
false
business_process_manager
false
business_process_user
false

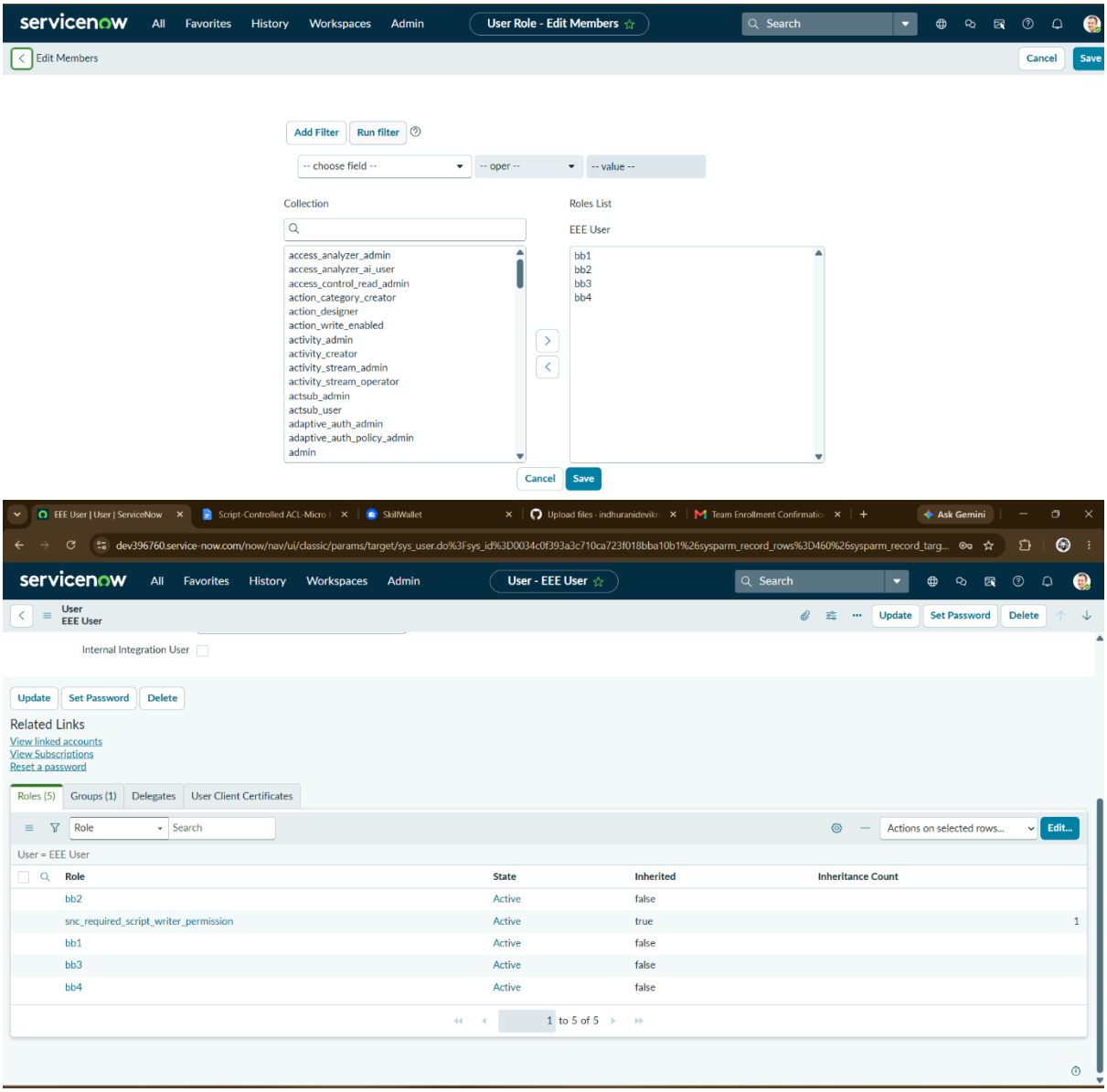

dev396760.service-now.com/now/nav/ui/classic/params/target/sys_user_role.do%3Fsys_id%3Df2d4883793a3c710ca723f018bba103d%26sysparm_record_target%3Dsys_user_role%26sysp...
servicenow
All
Favorites
History
Workspaces
Admin
Role - bb1
Search
Role
bb1
Update
Delete
Name
bb1
Application
Global
Elevated privilege
Description
Brief role purpose
Update
Delete
Contains Roles
Applications with Role
Modules with Role
Custom Tables
for text
Search
New
Edit...
Role = bb1
Contains


Role
bb2
Update
Delete
Name
bb2
Application
Global
Elevated privilege
Description
Brief role purpose
Update
Delete
Contains Roles
Applications with Role
Modules with Role
Custom Tables
for text
Search
New
Edit...
Role = bb2
Contains


dev396760.service-now.com/now/nav/ui/classic/params/target/sys_user_role.do%3Fsys_id%3D6c05c0f393a3c710ca723f018bba10ed%26sysparm_view%3D%26sysparm_record_row%3D3...
servicenow
All
Favorites
History
Workspaces
Admin
Role - bb3
Search
Role
bb3
Update
Delete
Name
bb3
Application
Global
Elevated privilege
Description
Brief role purpose
Update
Delete
Contains Roles
Applications with Role
Modules with Role
Custom Tables
for text
Search
New
Edit...
Role = bb3
Contains

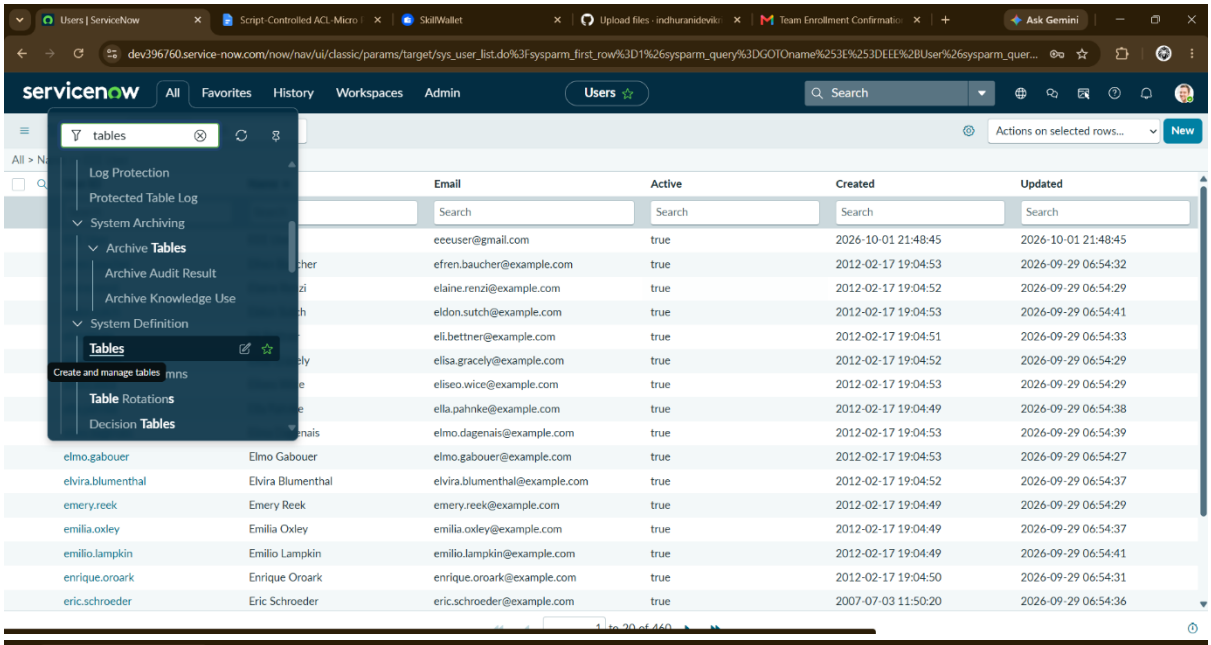

servicenow
All
Favorites
History
Workspaces
Admin
Role - bb4
Search
Role
bb4
Update
Delete
Name
bb4
Application
Global
Elevated privilege
Description
Brief role purpose
Update
Delete
Contains Roles
Applications with Role
Modules with Role
Custom Tables
for text
Search
New
Edit...
Role = bb4
Contains

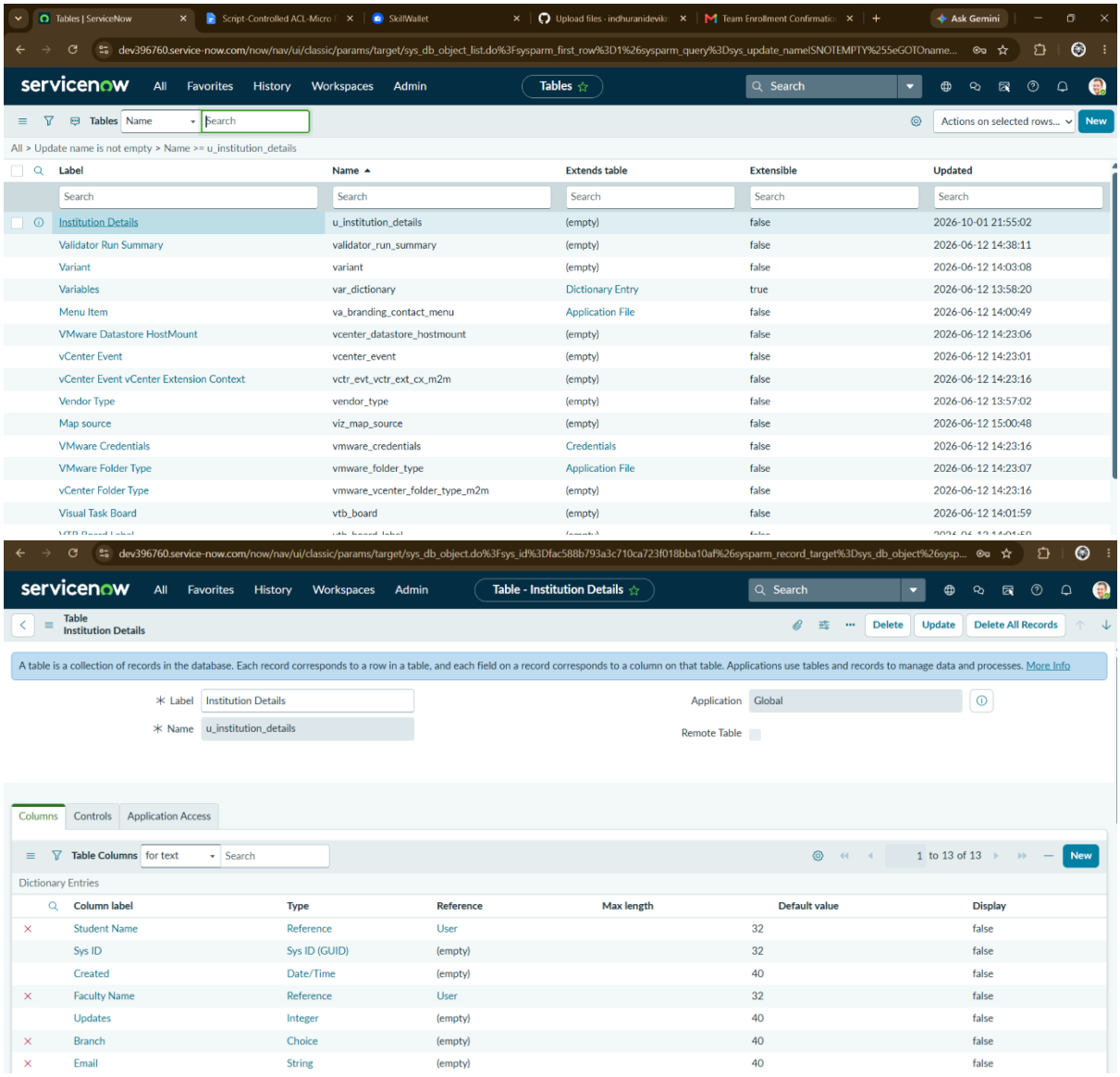

Users
Name
Search
Actions on selected rows...
New
All > Name >= EEE User
User ID
Name
Email
Active
Created
Updated
EEE User
EEE User
eeeuser@gmail.com
true
2026-10-01 21:48:45
2026-10-01 21:48:45
efren.baucher
Efren Baucher
efren.baucher@example.com
true
2012-02-17 19:04:53
2026-09-29 06:54:32
elaine.renzi
Elaine Renzi
elaine.renzi@example.com
true
2012-02-17 19:04:52
2026-09-29 06:54:29
eldon.sutch
Eldon Sutch
eldon.sutch@example.com
true
2012-02-17 19:04:53
2026-09-29 06:54:41
eli.bettner
Eli Bettner
eli.bettner@example.com
true
2012-02-17 19:04:51
2026-09-29 06:54:33
elisa.gracely
Elisa Gracely
elisa.gracely@example.com
true
2012-02-17 19:04:52
2026-09-29 06:54:29
eliseo.wice
Eliseo Wice
eliseo.wice@example.com
true
2012-02-17 19:04:53
2026-09-29 06:54:29
ella.pahnke
Ella Pahnke
ella.pahnke@example.com
true
2012-02-17 19:04:49
2026-09-29 06:54:38
elmo.dagenais
Elmo Dagenais
elmo.dagenais@example.com
true
2012-02-17 19:04:53
2026-09-29 06:54:39
elmo.gabouer
Elmo Gabouer
elmo.gabouer@example.com
true
2012-02-17 19:04:53
2026-09-29 06:54:27
elvira.blumenthal
Elvira Blumenthal
elvira.blumenthal@example.com
true
2012-02-17 19:04:52
2026-09-29 06:54:37
emery.reek
Emery Reek
emery.reek@example.com
true
2012-02-17 19:04:49
2026-09-29 06:54:29
emilia.oxley
Emilia Oxley
emilia.oxley@example.com
true
2012-02-17 19:04:49
2026-09-29 06:54:37



Role Creation

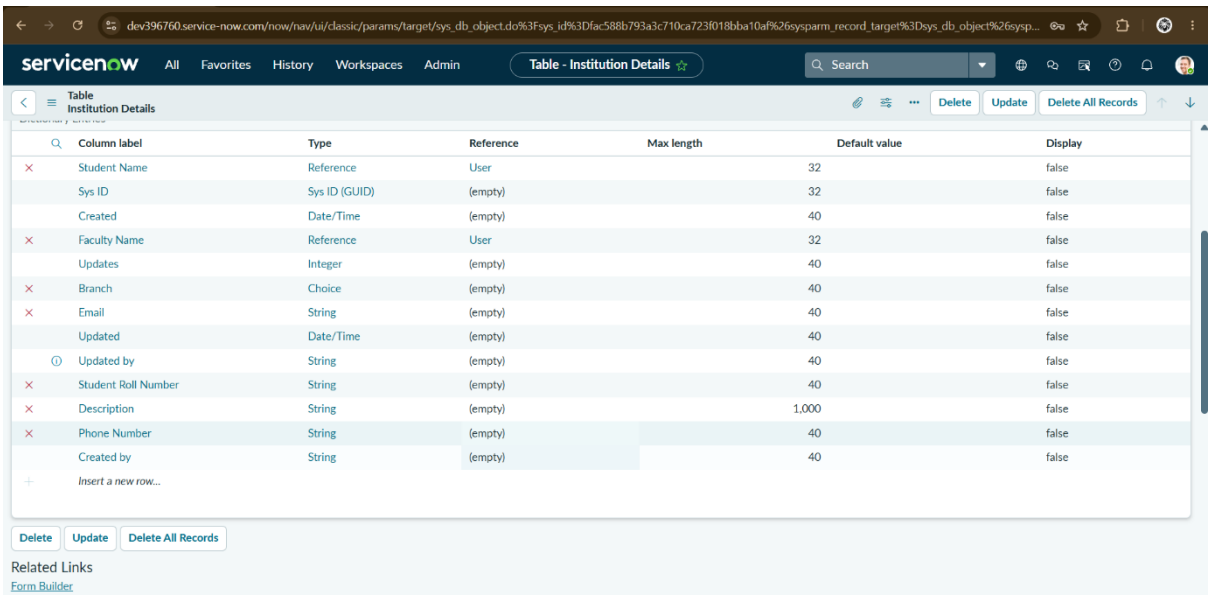
- Navigate to **User Administration → Roles**
- Click **New**
- **Name**: Enter role name as bb1
- **Description**: Brief role purpose
- Click **Submit**
- Create other roles also like **bb2,bb3,bb4.(follow the same procedure)**
- Assign all the roles to the EEE User

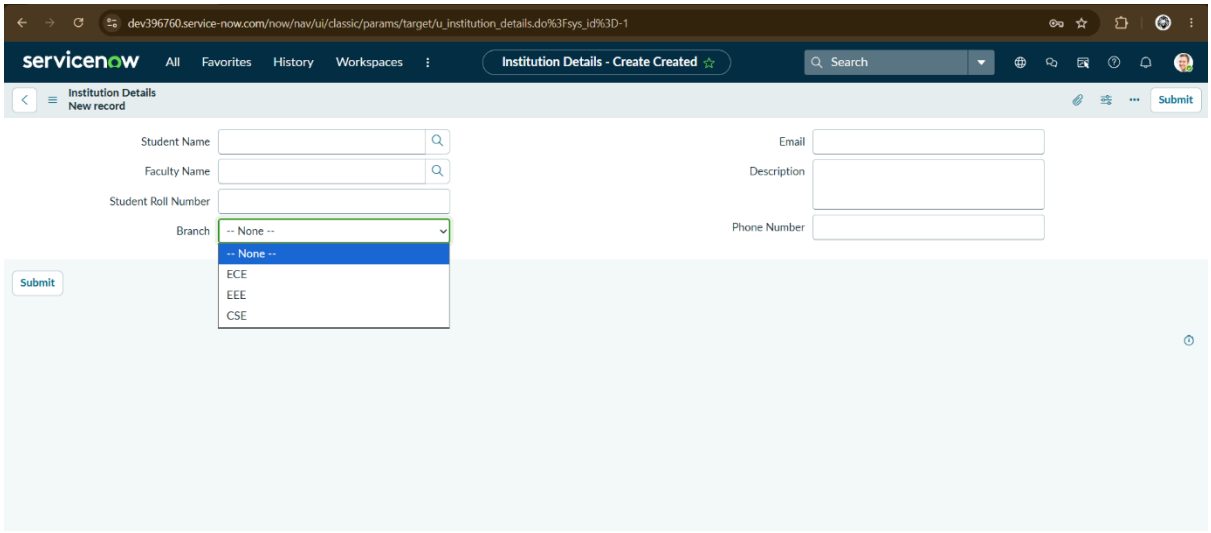
1. 

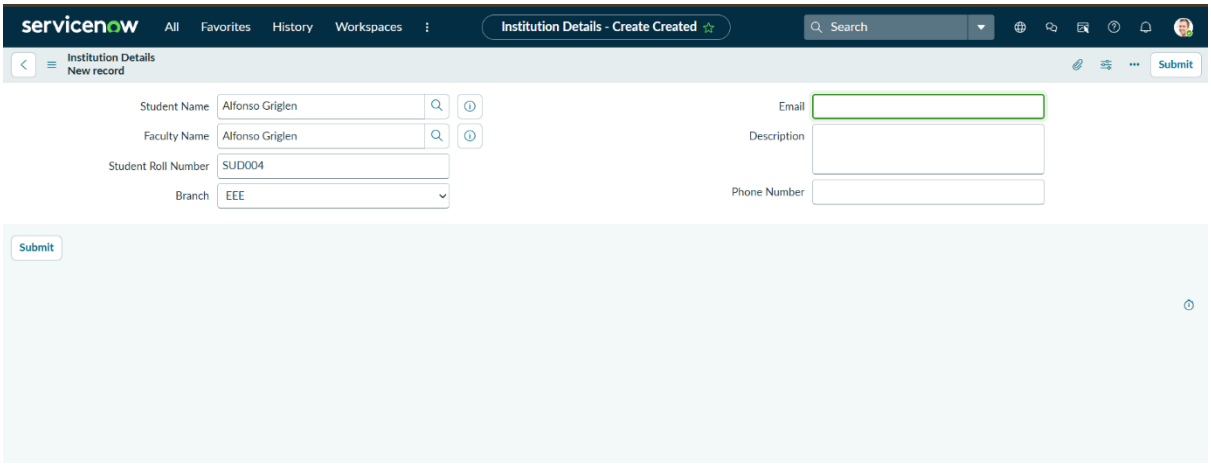


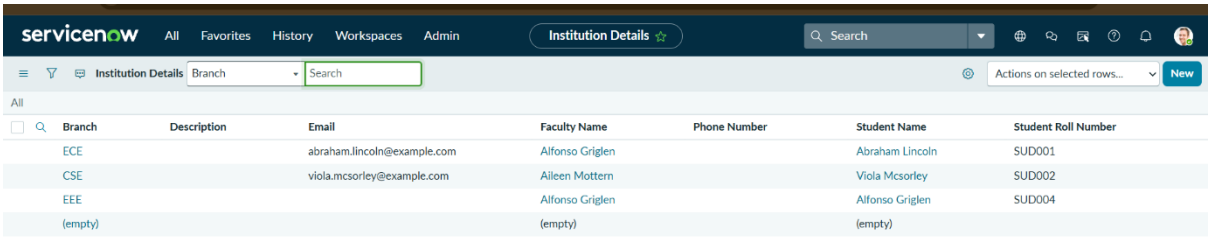
Click **Submit**.

2. Add the following fields to the **u_institution_details** table:
   - **Student Roll Number**– Auto Number
   - **Student Name** – Reference- User
   - **Faculty Name** – Reference- User
   - **Branch** – Choice (ECE, EEE, CSE)
   - **Email**– String
   - **Phone Number**– String
   - **Description**– Multi String
3. Save the Table
4. Create multiple records in the **Student Records** table with different branch values (ECE, EEE, CSE).









Creation of Access Control Lists(Read ACL)

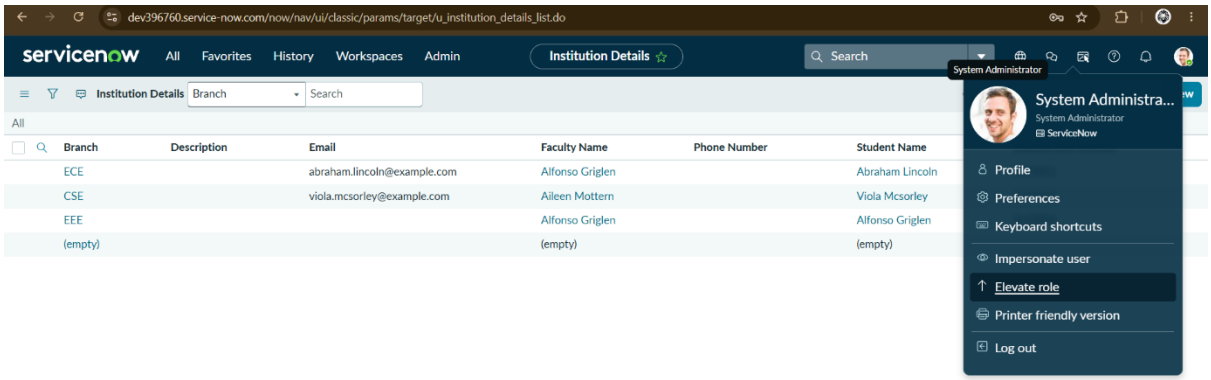
1. In the **Application Navigator**, search for **System Security → Access Control (ACL)**.
2. Elevate “Security_admin” role
3. Click **New** to create a new ACL.
4. Enter the following details:
   - **Type:** record
   - **Operation:** read
   - **Name:** u_institution_details
   - **Active:** true
   - **Advanced:** true
5. In the **Requires role** related list, add:

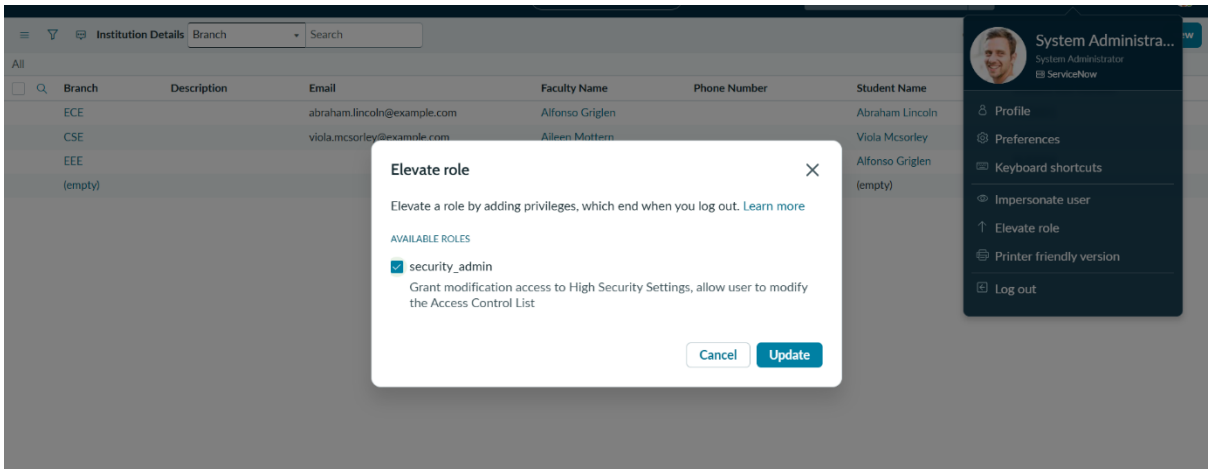
- bb1 (custom role)

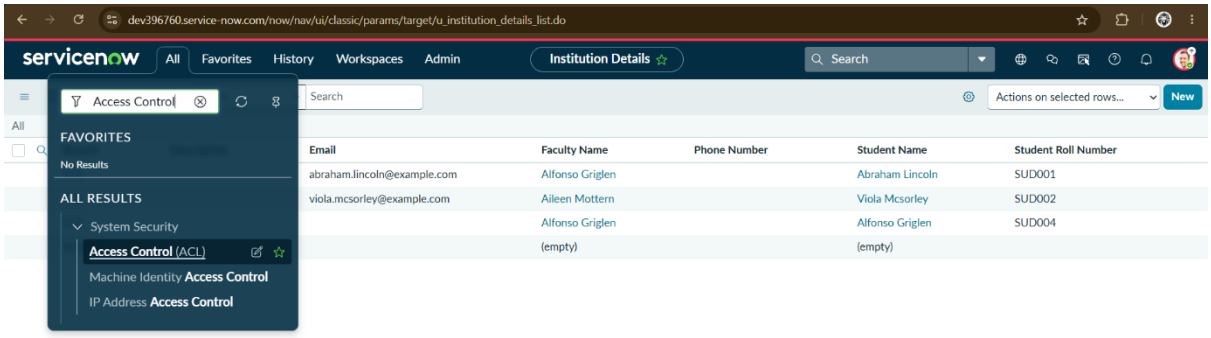
6. Add Data condition as **Branch is EEE**
    1. In the **Script** field, add the following script:

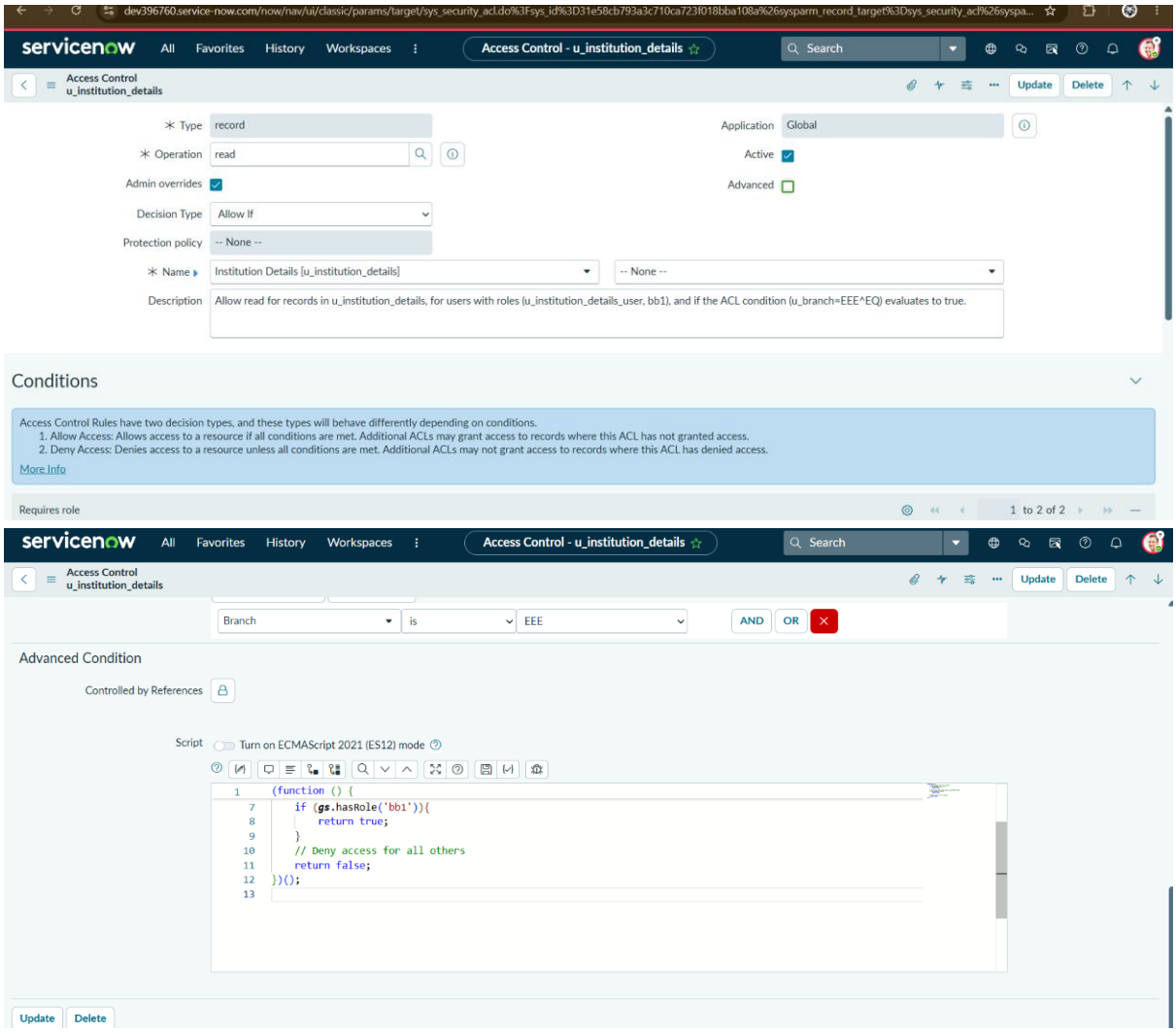
```
(function () {
   // Allow admin users full access
   if (gs.hasRole('admin')) {
      return true;
   }
   // Allow only EEE branch users to see EEE records
   if (gs.hasRole('bb1')){
      return true;
   }
   // Deny access for all others
   return false;
})();
```

2. Click **Submit** to save the ACL.





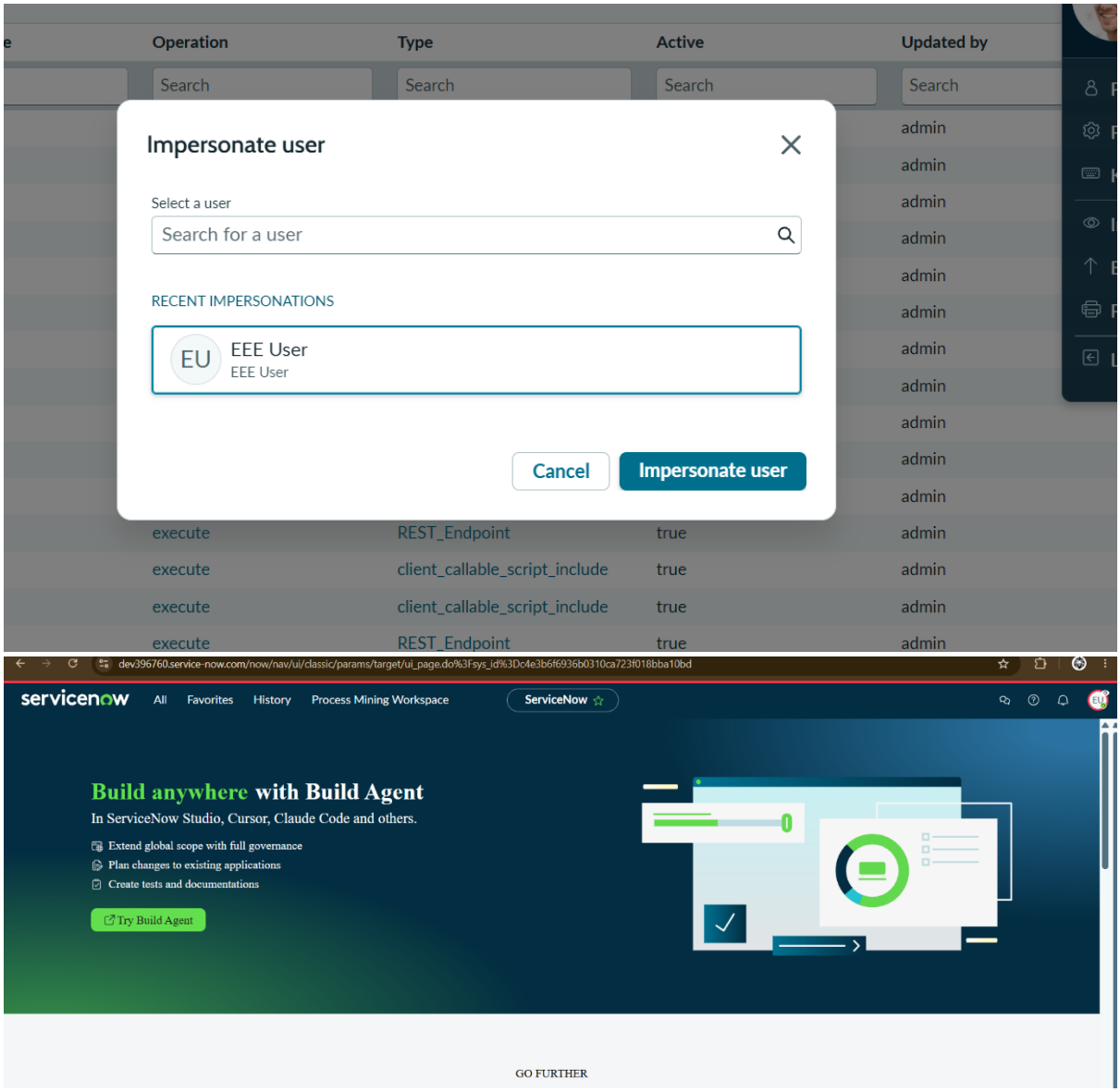




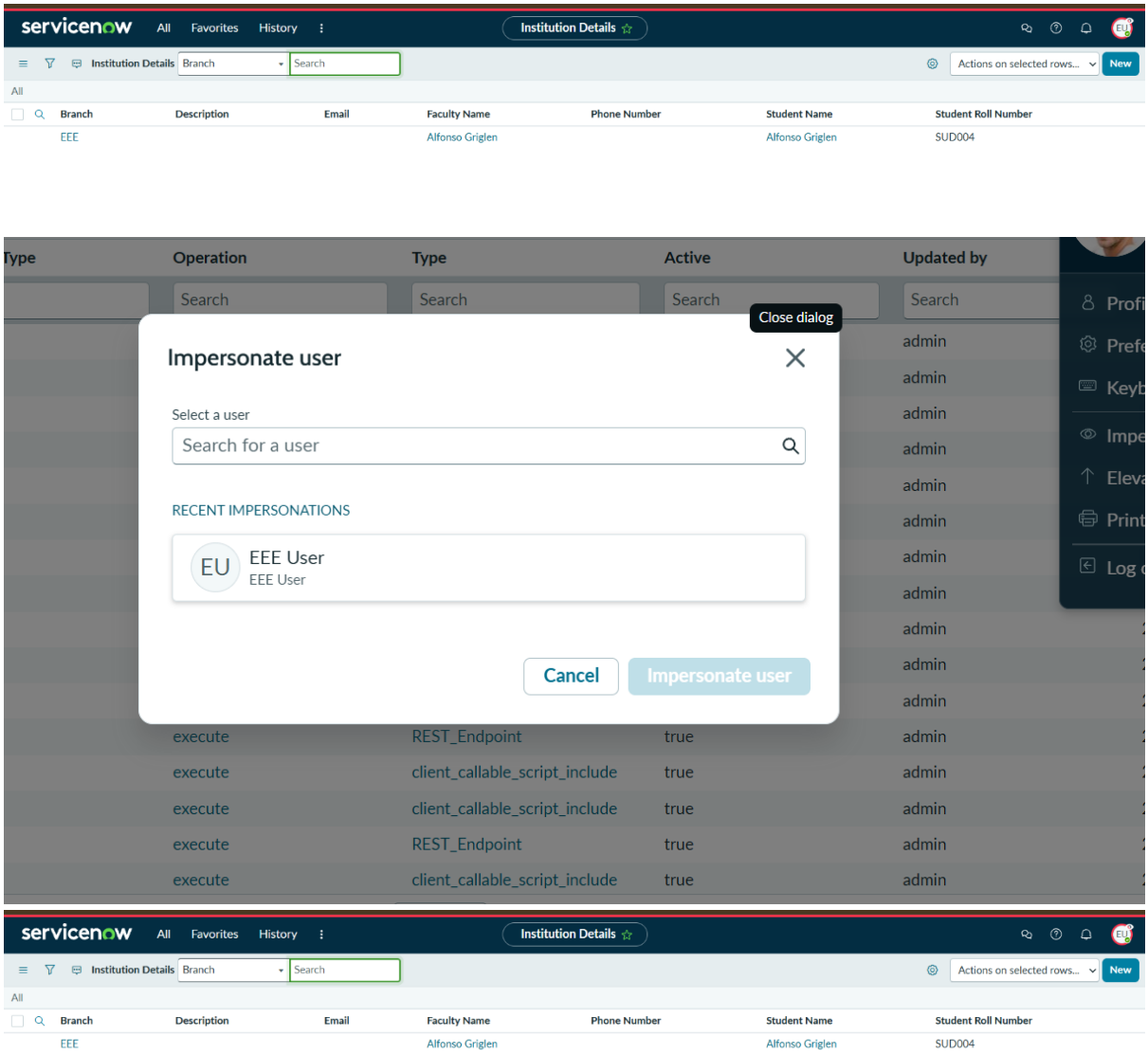
1. Impersonate **EEE users** and open the Student Records list.
2. Copy the table name u_institution_details.list/LIST and paste in the All menu.
3. Impersonate **Admin** and open the Student Records list.

Verify :

- User with **bb1 role** can view **only EEE branch records**

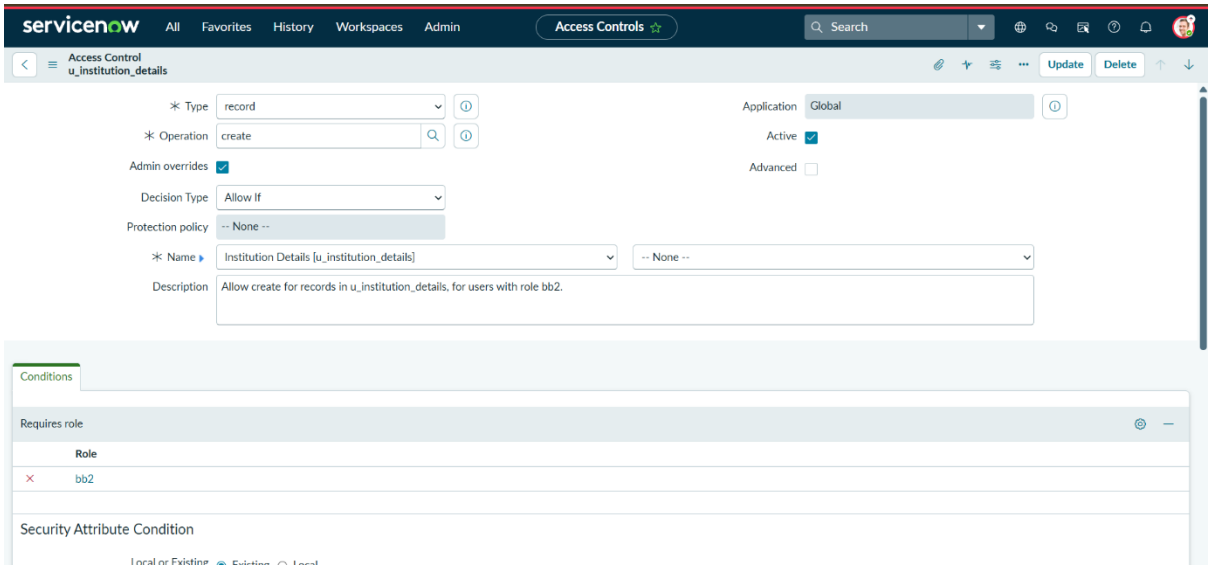


- User without the role cannot view any records
- Admin user can view **all records regardless of branch**



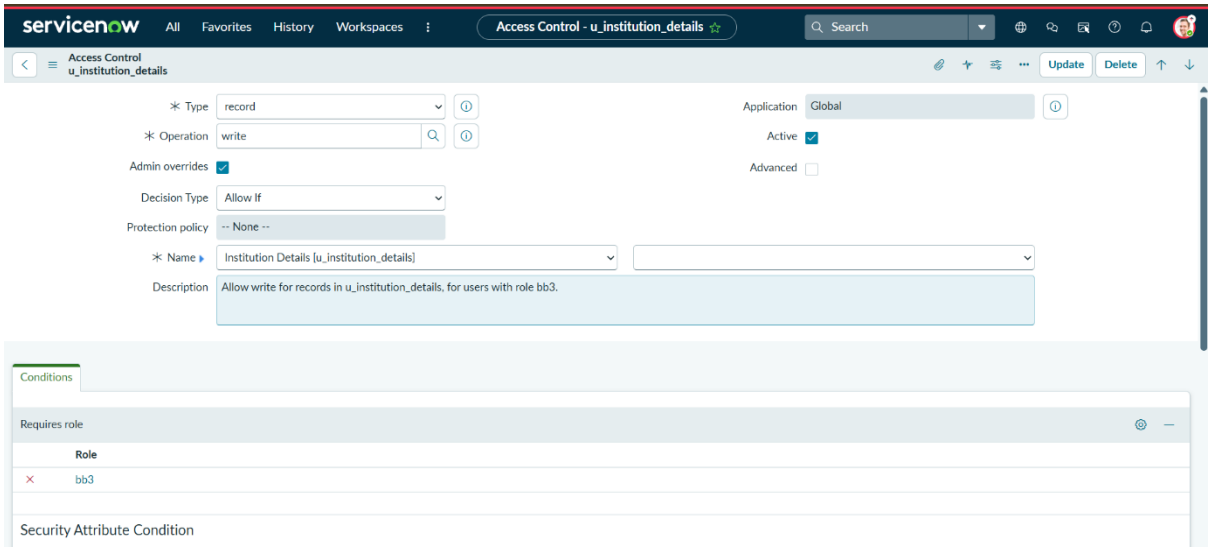
Creation of Access Control Lists(Create ACL)

1. Click **New** to create a new ACL(Create)
2. Enter the following details:
   - **Type:** record
   - **Operation:** create
   - **Name:** u_institution_details
   - **Active:** true
3. In the **Requires role** related list, add:
   - bb2 (custom role)
4. No need to add any Data condition
5. Save the ACL Record



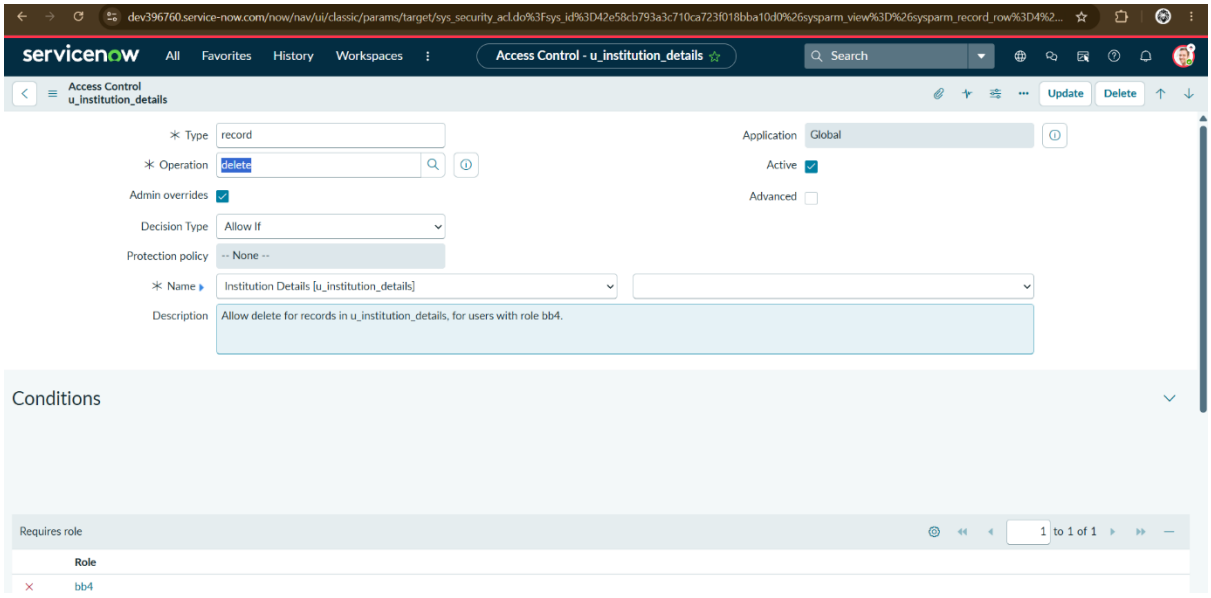
Creation of Access Control Lists(Write ACL)

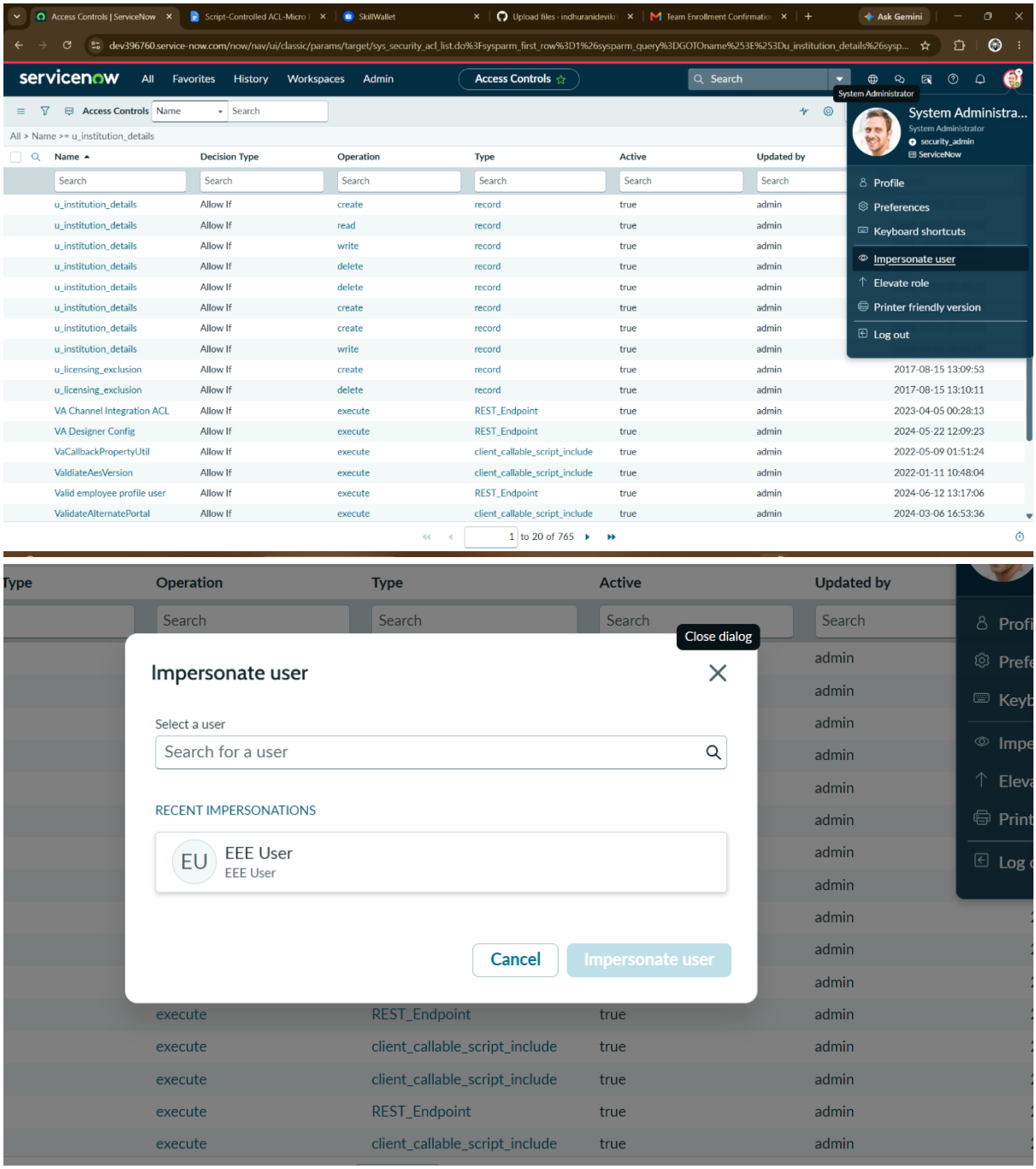
1. Click New to create a new ACL(Write)
2. Enter the following details:
   - Type: record
   - Operation: Write
   - Name: u_institution_details
   - Active: true
3. In the Requires role related list, add:
   - bb3 (custom role)
4. No need to add any Data condition
5. Save the ACL Record



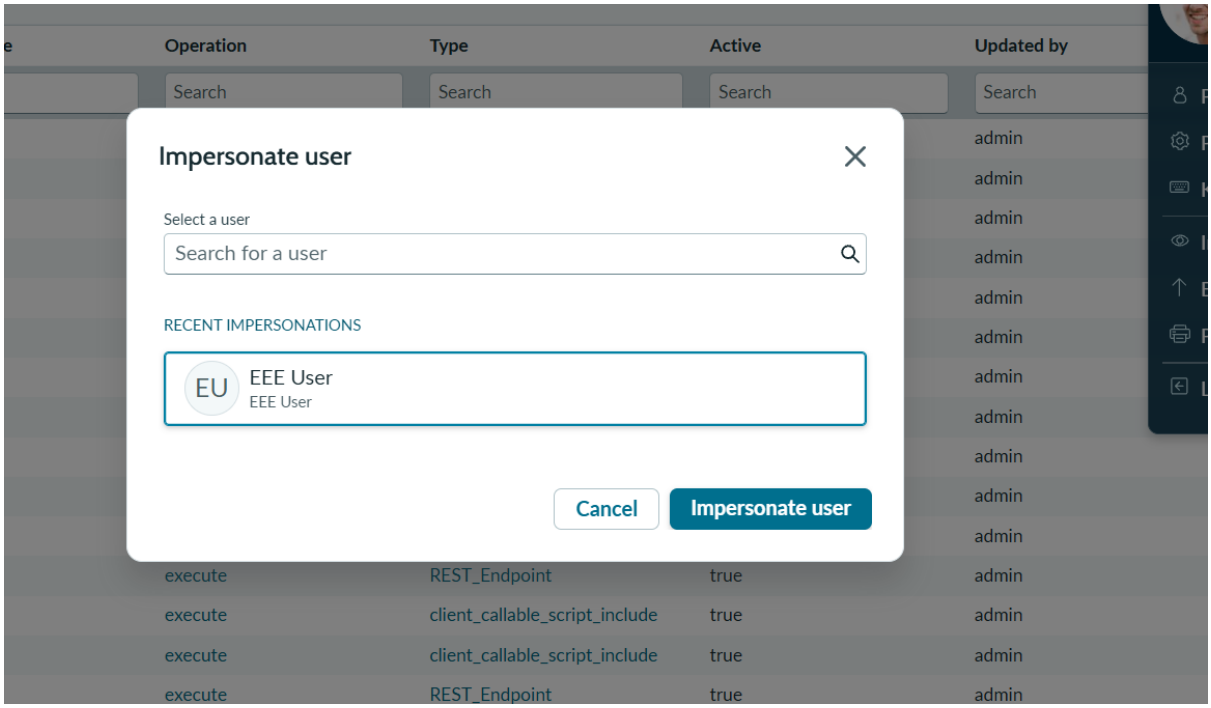
Creation of Access Control Lists(Write ACL)

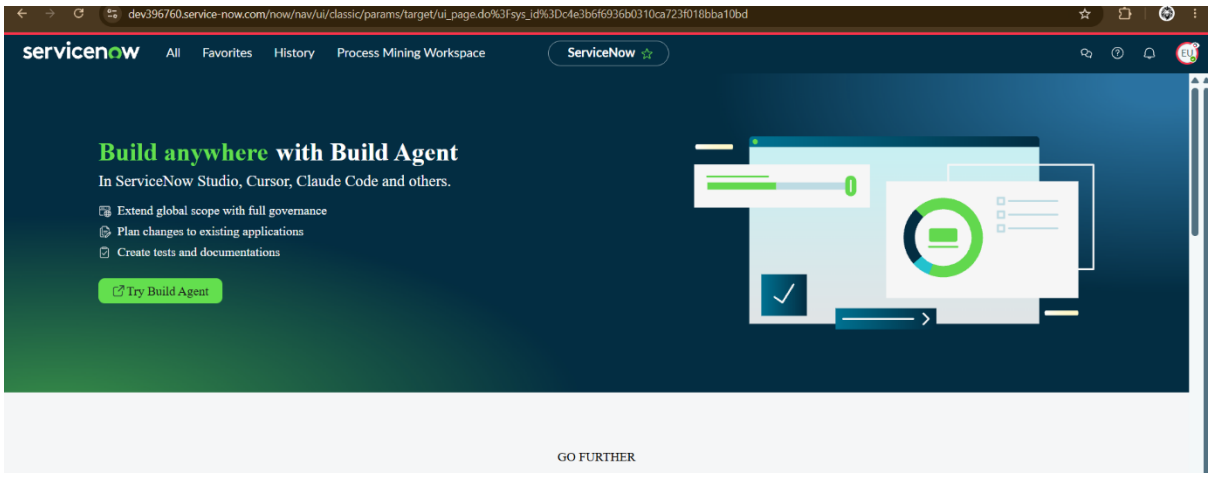
1. Click New to create a new ACL(Write)
2. Enter the following details:
   - Type: record
   - Operation: Write
   - Name: u_institution_details
   - Active: true
3. In the Requires role related list, add:
   - bb3 (custom role)
4. No need to add any Data condition
5. Save the ACL Record



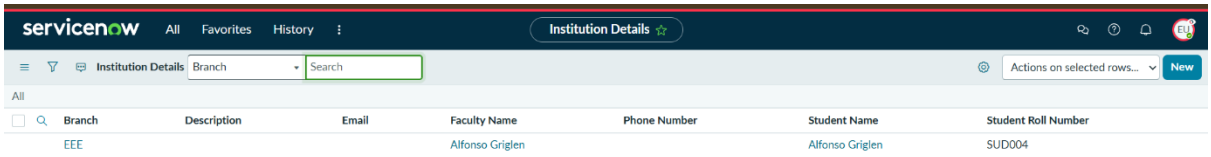

Access Controls | ServiceNow
Script-Controlled ACL-Micro
SkillWallet
Upload files · indhuranidevi
Team Enrollment Confirmation
Ask Gemini
dev396760.service-now.com/now/nav/ui/classic/params/target/sys_security_acl_list.do%3Fsysparm_first_row%3D1%26sysparm_query%3DGOTOname%253E%253Du_institution_details%26sysp...
servicenow
All
Favorites
History
Workspaces
Admin
Access Controls
Search
System Administrator
System Administra...
System Administrator
security_admin
ServiceNow
Profile
Preferences
Keyboard shortcuts
Impersonate user
Elevate role
Printer friendly version
Log out
Access Controls
Name
Search
All > Name >= u_institution_details
Name
Decision Type
Operation
Type
Active
Updated by
u_institution_details Allow If create record true admin
u_institution_details Allow If read record true admin
u_institution_details Allow If write record true admin
u_institution_details Allow If delete record true admin
u_institution_details Allow If delete record true admin
u_institution_details Allow If create record true admin
u_institution_details Allow If create record true admin
u_institution_details Allow If write record true admin
u_licensing_exclusion Allow If create record true admin 2017-08-15 13:09:53
u_licensing_exclusion Allow If delete record true admin 2017-08-15 13:10:11
VA Channel Integration ACL Allow If execute REST_Endpoint true admin 2023-04-05 00:28:13
VA Designer Config Allow If execute REST_Endpoint true admin 2024-05-22 12:09:23
VaCallbackPropertyUtil Allow If execute client_callable_script_include true admin 2022-05-09 01:51:24
ValdiateAesVersion Allow If execute client_callable_script_include true admin 2022-01-11 10:48:04
Valid employee profile user Allow If execute REST_Endpoint true admin 2024-06-12 13:17:06
ValidateAlternatePortal Allow If execute client_callable_script_include true admin 2024-03-06 16:53:36
1 to 20 of 765

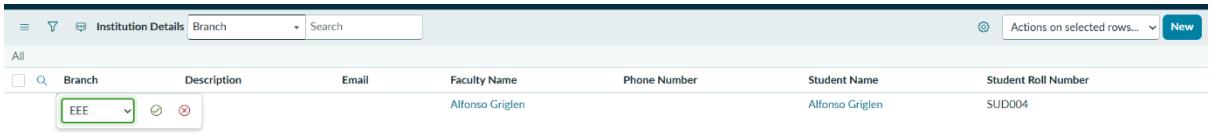
Type
Operation
Type
Active
Updated by
Search
Close dialog
Impersonate user
Select a user
Search for a user
RECENT IMPERSONATIONS
EU
EEE User
EEE User
Cancel
Impersonate user
admin
execute REST_Endpoint true admin
execute client_callable_script_include true admin
execute client_callable_script_include true admin
execute REST_Endpoint true admin
execute client_callable_script_include true admin

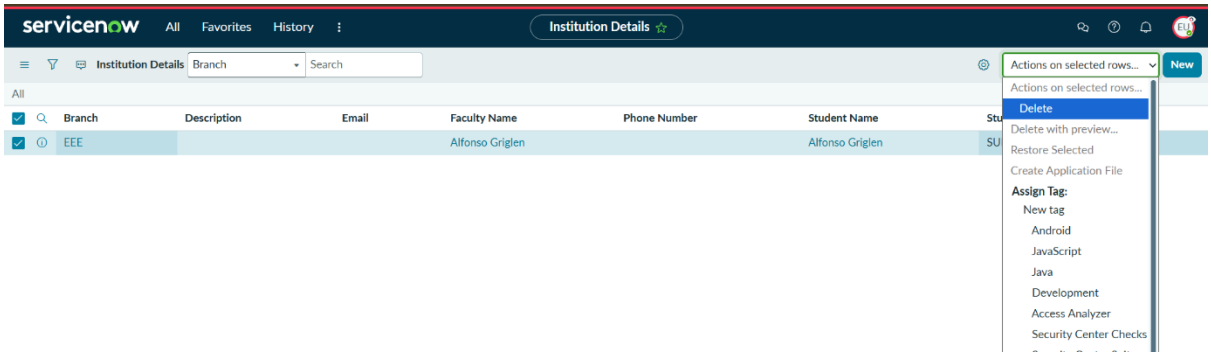

Operation
Type
Active
Updated by
Search
Impersonate user
Select a user
Search for a user
RECENT IMPERSONATIONS
EU
EEE User
EEE User
Cancel
Impersonate user
admin
execute
REST_Endpoint
true
client_callable_script_include

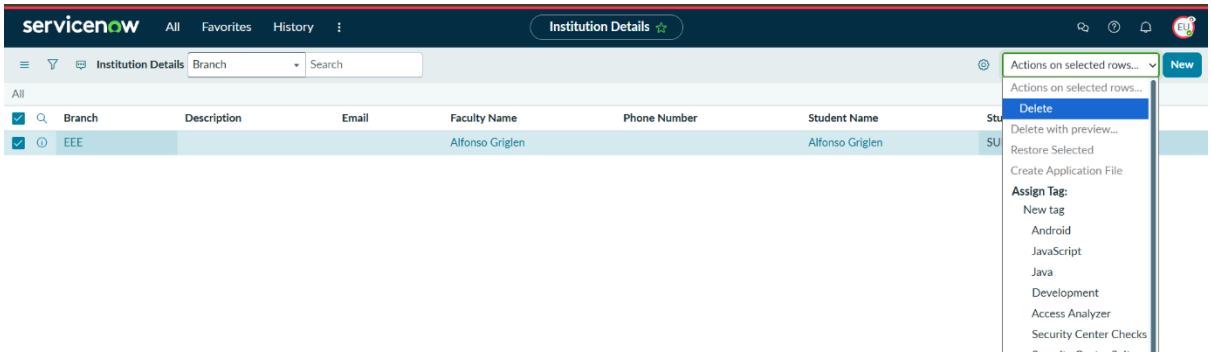

servicenow
All
Favorites
History
Process Mining Workspace
ServiceNow
Build anywhere with Build Agent
In ServiceNow Studio, Cursor, Claude Code and others.
Extend global scope with full governance
Plan changes to existing applications
Create tests and documentations
Try Build Agent

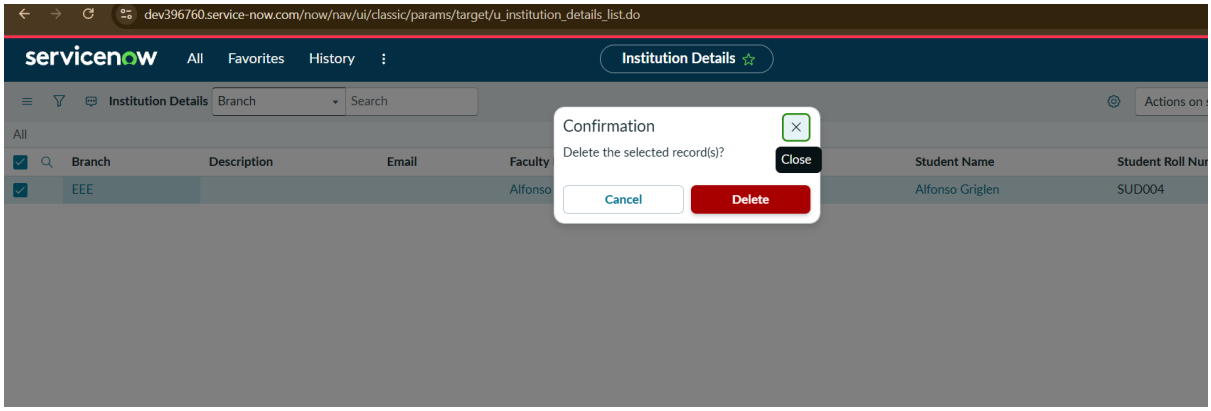
GO FURTHER

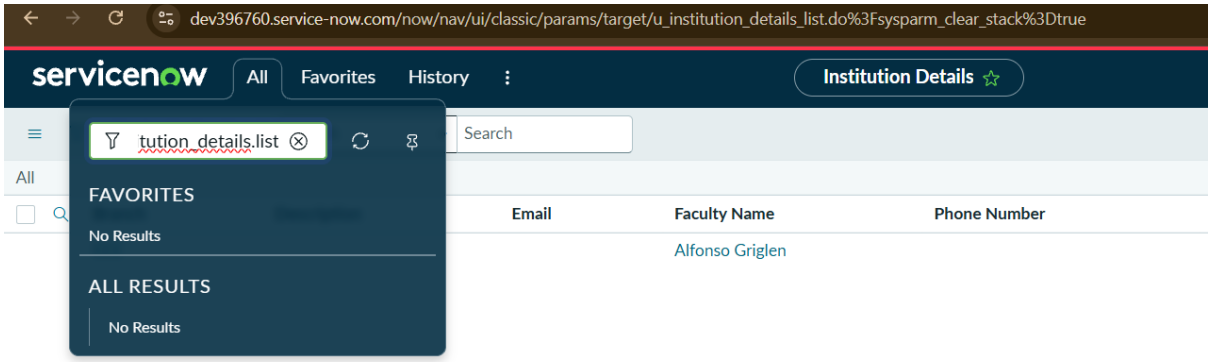

servicenow
All
Favorites
History
Institution Details
Branch
Search
Actions on selected rows...
New
Description
Email
Faculty Name
Phone Number
Student Name
Student Roll Number
EEE
Alfonso Griglen
Alfonso Griglen
SUD004

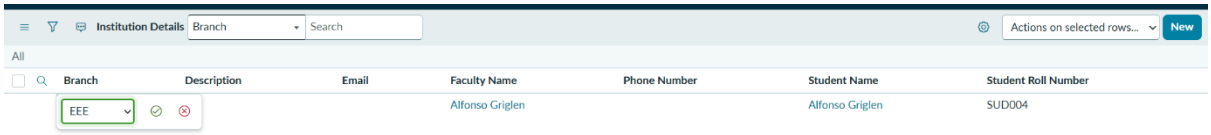

Institution Details
Branch
Search
Actions on selected rows...
New
All
Branch
Description
Email
Faculty Name
Phone Number
Student Name
Student Roll Number
EEE
Alfonso Griglen
Alfonso Griglen
SUD004


servicenow
All
Favorites
History
Institution Details
Institution Details
Branch
Search
Actions on selected rows...
New
All
Branch
Description
Email
Faculty Name
Phone Number
Student Name
EEE
Alfonso Griglen
Alfonso Griglen
Actions on selected rows...
Delete
Delete with preview...
Restore Selected
Create Application File
Assign Tag:
New tag
Android
JavaScript
Java
Development
Access Analyzer
Security Center Checks


servicenow
All
Favorites
History
Institution Details
Institution Details
Branch
Search
Actions on selected rows...
New
All
Branch
Description
Email
Faculty Name
Phone Number
Student Name
EEE
Alfonso Griglen
Alfonso Griglen
Actions on selected rows...
Delete
Delete with preview...
Restore Selected
Create Application File
Assign Tag:
New tag
Android
JavaScript
Java
Development
Access Analyzer
Security Center Checks


dev396760.service-now.com/now/nav/ui/classic/params/target/u_institution_details_list.do
servicenow
All
Favorites
History
Institution Details
Institution Details
Branch
Search
All
Branch
Description
Email
Student Name
Student Roll Nu
EEE
Alfonso Griglen
SUD004
Confirmation
Delete the selected record(s)?
Close
Cancel
Delete





Conclusion :

The Script-Controlled ACL – Restrict Record Access Based on Field Value project successfully demonstrates how ServiceNow ACLs and scripting can be used to implement secure record-level access control. The project controls Read, Create, Write, and Delete operations using different user roles and field-based conditions. The system allows authorized users to access the required EEE branch records while restricting unauthorized users, and administrators retain full access. Through this project, we gain a clear understanding of how user roles, record field values, and ACL scripts work together to protect sensitive data and prevent unauthorized access. Overall, the project provides a practical approach to implementing controlled and secure record access within the ServiceNow platform.